MATH 1210 Worksheet 6 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

`http://home.cc.umanitoba.ca/~thomas/Courses`

1. Consider the matrix

$$A = \begin{pmatrix} 2i & 0 & \sqrt{2} \\ 0 & -\sqrt{2} & 1 \\ 1 & -i & \sqrt{2} \end{pmatrix}^4 .$$

Compute the determinant of A. *Hint: Use that* $|AB| = |A||B|$

Solution: First we compute the determinant

$$\begin{vmatrix} 2i & 0 & \sqrt{2} \\ 0 & -\sqrt{2} & 1 \\ 1 & -i & \sqrt{2} \end{vmatrix} = 2i \begin{vmatrix} -\sqrt{2} & 1 \\ -i & \sqrt{2} \end{vmatrix} + \begin{vmatrix} 0 & \sqrt{2} \\ -\sqrt{2} & 1 \end{vmatrix} = 2i(-2+i) + 2 = -4i$$

where we expanded along the first column. Hence, using the hint we find that

$$|A| = \left| \begin{pmatrix} 2i & 0 & \sqrt{2} \\ 0 & -\sqrt{2} & 1 \\ 1 & -i & \sqrt{2} \end{pmatrix}^4 \right| = \begin{vmatrix} 2i & 0 & \sqrt{2} \\ 0 & -\sqrt{2} & 1 \\ 1 & -i & \sqrt{2} \end{vmatrix}^4 = (-4i)^4 = 256.$$

2. Use Gaussian elimination to compute the determinant of the matrix

$$A = \begin{pmatrix} 2 & 3 & 2 & -1 & -2 \\ 0 & -1 & -1 & 1 & 0 \\ -3 & 1 & 1 & 2 & 1 \\ 1 & -2 & 2 & 1 & -1 \\ 0 & 1 & 1 & 2 & -1 \end{pmatrix}$$

Solution: We can simplify the computation of the determinant by using the following sequence of steps

$$|A| = \begin{vmatrix} 2 & 3 & 2 & -1 & -2 \\ 0 & -1 & -1 & 1 & 0 \\ -3 & 1 & 1 & 2 & 1 \\ 1 & -2 & 2 & 1 & -1 \\ 0 & 1 & 1 & 2 & -1 \end{vmatrix} \begin{matrix} R_1 \to R_1 - 2R_4 \\ \\ R_3 \to R_3 + 3R_4 \\ \\ \\ \end{matrix} = \begin{vmatrix} 0 & 7 & -2 & -3 & 0 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & -5 & 7 & 5 & -2 \\ 1 & -2 & 2 & 1 & -1 \\ 0 & 1 & 1 & 2 & -1 \end{vmatrix}.$$

By expanding along the first column we get

$$|A| = -\begin{vmatrix} 7 & -2 & -3 & 0 \\ -1 & -1 & 1 & 0 \\ -5 & 7 & 5 & -2 \\ 1 & 1 & 2 & -1 \end{vmatrix} \begin{matrix} \\ \\ R_3 \to R_3 - 2R_4 \\ \\ \end{matrix} = -\begin{vmatrix} 7 & -2 & -3 & 0 \\ -1 & -1 & 1 & 0 \\ -7 & 5 & 1 & 0 \\ 1 & 1 & 2 & -1 \end{vmatrix}.$$

By expanding along the last column we get

$$|A| = \begin{vmatrix} 7 & -2 & -3 \\ -1 & -1 & 1 \\ -7 & 5 & 1 \end{vmatrix}_{R_3 \to R_3 + R_1} = \begin{vmatrix} 7 & -2 & -3 \\ -1 & -1 & 1 \\ 0 & 3 & -2 \end{vmatrix}^{R_1 \to R_1 + 7R_2} = \begin{vmatrix} 0 & -9 & 4 \\ -1 & -1 & 1 \\ 0 & 3 & -2 \end{vmatrix}.$$

Expanding along the first column finally yields

$$|A| = \begin{vmatrix} -9 & 4 \\ 3 & -2 \end{vmatrix} = (-9) \cdot (-2) - 3 \cdot 4 = 18 - 12 = 6.$$

3. For a real number x consider the matrix

$$A = \begin{pmatrix} 1 & x & 2 \\ x & 0 & 2 \\ 1 & 1 & 1+x \end{pmatrix}.$$

(a) Compute the determinant of A for $x = 1$.

Solution: For $x = 1$ the matrix is

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 2 \\ 1 & 1 & 2 \end{pmatrix}.$$

We see that the first and the last row are the same, hence $|A| = 0$ when $x = 1$.

(b) Find all values of x for which $|A| = 0$. *Hint: use part (a).*

Solution: We compute the determinant of A by expanding along the second row:

$$\begin{aligned} |A| &= -x \begin{vmatrix} x & 2 \\ 1 & 1+x \end{vmatrix} - 2 \begin{vmatrix} 1 & x \\ 1 & 1 \end{vmatrix} \\ &= -x(x^2 + x - 2) - 2(1 - x) \\ &= -x^3 - x^2 + 4x - 2. \end{aligned}$$

Hence we must solve the equation

$$x^3 + x^2 - 4x + 2 = 0. \tag{1}$$

By part (a) we know that $x = 1$ is a solution of equation (2). Thus $x - 1$ is a factor of $x^3 + x^2 - 4x + 2$. Polynomial division gives

$$\begin{array}{rl} & x^2 + 2x - 2. \\ x - 1 \big) & x^3 + x^2 - 4x + 2 \\ & \underline{-x^3 + x^2} \\ & 2x^2 - 4x \\ & \underline{-2x^2 + 2x} \\ & -2x + 2 \\ & \underline{2x - 2} \\ & 0 \end{array}$$

Hence

$$x^3 + x^2 - 4x + 2 = (x - 1)(x^2 + 2x - 2).$$

The roots of the polynomial $x^2 + 2x - 2$ are

$$x = \frac{-2 \pm \sqrt{2^2 - 4(-2)}}{2} = -1 \pm \sqrt{3}.$$

In conclusion, for

$$x = 1, \quad x = -1 + \sqrt{3} \quad \text{or} \quad x = -1 - \sqrt{3}$$

we have $|A| = 0$.

4. Use Cramer's rule to solve the system

$$\begin{aligned} 2x - 3y + 6z &= 1 \\ 3x + y - z &= 0 \\ x + y - 2z &= -1. \end{aligned}$$

Solution: The system can be written in matrix form $AX = B$ where

$$A = \begin{pmatrix} 2 & -3 & 6 \\ 3 & 1 & -1 \\ 1 & 1 & -2 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

By expanding along the first row we find that the determinant of A is

$$\begin{vmatrix} 2 & -3 & 6 \\ 3 & 1 & -1 \\ 1 & 1 & -2 \end{vmatrix} = 2\begin{vmatrix} 1 & -1 \\ 1 & -2 \end{vmatrix} - 3\begin{vmatrix} -3 & 6 \\ 1 & -2 \end{vmatrix} + \begin{vmatrix} -3 & 6 \\ 1 & -1 \end{vmatrix} = -5.$$

It is nonzero and by Cramer's rule the system has unique solution. Moreover, by Cramer's rule the solution is (x, y, z) where:

$$x = \frac{|A_1|}{|A|} = -\frac{\begin{vmatrix} 1 & -3 & 6 \\ 0 & 1 & -1 \\ -1 & 1 & -2 \end{vmatrix}}{5} = -\frac{\begin{vmatrix} 1 & -1 \\ 1 & -2 \end{vmatrix} - \begin{vmatrix} -3 & 6 \\ 1 & -1 \end{vmatrix}}{5} = -\frac{2}{5}$$

$$y = \frac{|A_2|}{|A|} = -\frac{\begin{vmatrix} 2 & 1 & 6 \\ 3 & 0 & -1 \\ 1 & -1 & -2 \end{vmatrix}}{5} = -\frac{2\begin{vmatrix} 0 & -1 \\ -1 & -2 \end{vmatrix} - 3\begin{vmatrix} 1 & 6 \\ -1 & -2 \end{vmatrix} + \begin{vmatrix} 1 & 6 \\ 0 & -1 \end{vmatrix}}{5} = \frac{15}{5} = 3$$

$$z = \frac{|A_3|}{|A|} = -\frac{\begin{vmatrix} 2 & -3 & 1 \\ 3 & 1 & 0 \\ 1 & 1 & -1 \end{vmatrix}}{5} = -\frac{2\begin{vmatrix} 1 & 0 \\ 1 & -1 \end{vmatrix} - 3\begin{vmatrix} -3 & 1 \\ 1 & -1 \end{vmatrix} + \begin{vmatrix} -3 & 1 \\ 1 & 0 \end{vmatrix}}{5} = \frac{9}{5}.$$

Note that in the three cases we expanded the determinant along the first column.

5. Consider the system

$$\begin{aligned} ax_1 + 6x_3 &= 12 \\ x_1 + 3x_2 + 4x_3 &= -1 \\ x_2 + x_3 &= b, \end{aligned}$$

where a and b are two real numbers.

(a) For which value(s) of a and b does the system have a unique solution?

Solution: The system can be written as $AX = B$ where

$$A = \begin{pmatrix} a & 0 & 6 \\ 1 & 3 & 4 \\ 0 & 1 & 1 \end{pmatrix}.$$

and

$$B = \begin{pmatrix} 12 \\ -1 \\ b \end{pmatrix}.$$

By expanding along the first row we find that the determinant of A is

$$\begin{vmatrix} a & 0 & 6 \\ 1 & 3 & 4 \\ 0 & 1 & 1 \end{vmatrix} = a \begin{vmatrix} 3 & 4 \\ 1 & 1 \end{vmatrix} + 6 \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = -a + 6.$$

The system has a unique solution if and only if when the determinant of A is nonzero. This is the case when $a \neq 6$, and b can be arbitrary.

(b) For which value(s) of a and b does the system have at least two solutions?

Solution: By part (a), the system has a unique solution if and only if $a \neq 6$. Hence, in order to have at least two solutions, we must have $a = 6$. If we substitute $a = 6$, the system becomes

$$\begin{aligned} 6x_1 + 6x_3 &= 12 \\ x_1 + 3x_2 + 4x_3 &= -1 \\ x_2 + x_3 &= b, \end{aligned}$$

and the corresponding augmented matrix is

$$\left(\begin{array}{ccc|c} 6 & 0 & 6 & 12 \\ 1 & 3 & 4 & -1 \\ 0 & 1 & 1 & b \end{array}\right)$$

There are two possible solutions: either the system is inconsistent (no solutions) or it has infinitely many solutions. To find out, we perform Gauss-Jordan elimination:

$$\begin{aligned} \left(\begin{array}{ccc|c} 6 & 0 & 6 & 12 \\ 1 & 3 & 4 & -1 \\ 0 & 1 & 1 & b \end{array}\right) R_1 \to R_1/6 &\longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 1 & 3 & 4 & -1 \\ 0 & 1 & 1 & b \end{array}\right) R_2 \to R_2 - R_1 \\ &\longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 3 & 3 & -3 \\ 0 & 1 & 1 & b \end{array}\right) R_2 \to R_2/3 \\ &\longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & b+1 \end{array}\right) R_3 \to R_3 - R_2. \end{aligned}$$

If $b \neq -1$, the system is inconsistent and has no solutions. For $b = -1$, the augmented matrix is in reduced row echelon form:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right),$$

and the system has infinitely many solutions. In conlusion, the system has at least two solutions if $a = 6$ and $b = -1$.

6. Consider the system

$$\begin{aligned} 2x + z &= 3 \\ x + 2y + tz &= 5 \\ -x + 2y &= 4 \end{aligned}$$

where t is a real number. It is given to you that the system has a unique solution (x, y, z) with $z = 1$. Use Cramer's rule to find t.

Solution: The coefficient matrix of the system is

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 2 & t \\ -1 & 2 & 0 \end{pmatrix}.$$

If (x, y, z) is a solution of the system, then by Cramer's rule we have

$$z = \frac{|A_3|}{|A|}$$

where

$$A_3 = \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 5 \\ -1 & 2 & 4 \end{pmatrix}.$$

It is given to us that $z = 1$ so we have

$$\begin{aligned} 1 &= \frac{|A_3|}{|A|} \\ \iff |A| &= |A_3|. \end{aligned} \tag{2}$$

By expanding along the first row we find that the determinant of A is

$$\begin{vmatrix} 2 & 0 & 1 \\ 1 & 2 & t \\ -1 & 2 & 0 \end{vmatrix} = 2 \begin{vmatrix} 2 & t \\ 2 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} = -4t + 4.$$

By expanding along the first row again we also find that the determinant of A_3 is

$$\begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 5 \\ -1 & 2 & 4 \end{pmatrix} = 2 \begin{vmatrix} 2 & 5 \\ 2 & 4 \end{vmatrix} + 3 \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} = 8.$$

Substitution in (2) gives the equation

$$8 = -4t + 4.$$

We solve for t and find

$$t = -1.$$

7. Determine if each of the following set of vectors are linearly independent.

(a) $\mathbf{u}_1 = \langle 1, -3, 7\rangle$, $\mathbf{u}_2 = \langle 5, 2, 3\rangle$, $\mathbf{u}_3 = \langle 13, -7, -1\rangle$, $\mathbf{u}_4 = \langle 1, 1, -1\rangle$.

Solution: The set contains $m = 4$ vectors with $n = 3$ components. Since $m > n$ the vectors must be linearly dependent.

(b) $\mathbf{u}_1 = \langle 2, 0, 3\rangle$, $\mathbf{u}_2 = \langle 1, -1, 3\rangle$, $\mathbf{u}_3 = \langle 6, -2, 2\rangle$.

Solution: We compute the determinant by expanding along the first column

$$\begin{vmatrix} 2 & 1 & 6 \\ 0 & -1 & -2 \\ 3 & 3 & 2 \end{vmatrix} = 2\begin{vmatrix} -1 & -2 \\ 3 & 2 \end{vmatrix} + 3\begin{vmatrix} 1 & 6 \\ -1 & -2 \end{vmatrix} = 20.$$

Since it is nonzero the vectors are linearly independent.

8. (a) Are the following three vectors

$$\mathbf{u}_1 = \langle 1, 5, 2, 2\rangle, \quad \mathbf{u}_2 = \langle 2, 4, 1, 2\rangle, \quad \mathbf{u}_3 = \langle -1, 1, 0, -1\rangle$$

linearly dependent?

Solution: Determining if the three vectors are linearly dependent amounts to determine if the equation

$$C_1\mathbf{u}_1 + C_2\mathbf{u}_2 + C_3\mathbf{u}_3 = 0 \tag{3}$$

has a non-trivial solution (C_1, C_2, C_3). After writing the components of the vectors equation (3) becomes

$$C_1\langle 1, 5, 2, 2\rangle + C_2\langle 2, 4, 1, 2\rangle + C_3\langle -1, 1, 0, -1\rangle = 0.$$

Since the equations holds componentwise we can write the equation as a homogeneous system of linear equations

$$\begin{aligned} C_1 + 2C_2 - C_3 &= 0 \\ 5C_1 + 4C_1 + C_3 &= 0 \\ 2C_1 + C_2 &= 0 \\ 2C_1 + 2C_2 - C_3 &= 0. \end{aligned}$$

We find the reduced row echelon form by performing the following sequence of elementary row operations

$$
\begin{aligned}
\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 5 & 4 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 2 & 2 & -1 & 0 \end{array}\right)
\begin{array}{l} \\ R_2 \to R_2 - 5R_1 \\ R_3 \to R_3 - 2R_1 \\ R_4 \to R_4 - 2R_1 \end{array}
&\longrightarrow
\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -6 & 6 & 0 \\ 0 & -3 & 2 & 0 \\ 0 & -2 & 1 & 0 \end{array}\right)
\begin{array}{l} \\ R_2 \to -R_2/6 \\ \\ \\ \end{array} \\
&\longrightarrow
\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -3 & 2 & 0 \\ 0 & -2 & 1 & 0 \end{array}\right)
\begin{array}{l} R_1 \to R_1 - 2R_2 \\ \\ R_3 \to 3R_2 + R_3 \\ R_4 \to R_4 + 2R_2 \end{array} \\
&\longrightarrow
\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right)
\begin{array}{l} R_1 \to R_1 + R_3 \\ R_2 \to R_2 - R_3 \\ \\ R_4 \to R_4 - R_3 \end{array} \\
&\longrightarrow
\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)
\begin{array}{l} \\ \\ R_3 \to -R_3 \\ \\ \end{array} \\
&\longrightarrow
\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right).
\end{aligned}
$$

Hence $C_1 = C_2 = C_3 = 0$ and the vectors $\mathbf{u}_1$,$\mathbf{u}_2$,$\mathbf{u}_3$ are linearly independent.

(b) Write the vector $\mathbf{v} = \langle 3, 3, -1, 1\rangle$ as a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$. Justify your answer.

Solution: Writing $\mathbf{v}$ as a linear combination of $\mathbf{u}_1$,$\mathbf{u}_2$,$\mathbf{u}_3$ requires to find C_1, C_2, C_3 such that

$$C_1\mathbf{u}_1 + C_2\mathbf{u}_2 + C_3\mathbf{u}_3 = \mathbf{v}. \tag{4}$$

After writing the components of the vectors, the equation (4) becomes

$$C_1\langle 1, 5, 2, 2\rangle + C_2\langle 2, 4, 1, 2\rangle + C_3\langle -1, 1, 0, -1\rangle = \langle 3, 3, -1, 1\rangle.$$

Since the equations must hold componentwise, we can write the equation as a nonhomogeneous system

$$
\begin{aligned}
C_1 + 2C_2 - C_3 &= 3 \\
5C_1 + 4C_1 + C_3 &= 3 \\
2C_1 + C_2 &= -1 \\
2C_1 + 2C_2 - C_3 &= 1.
\end{aligned}
$$

Let us now solve the system to determine the solution (C_1, C_2, C_3), if it exists. The coefficient matrix of the system is the same as in part (a), with a nonzero vector on the righthandside. Hence we can do the same elementary row operations and only have to compute the effect on the last column:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 5 & 4 & 1 & 3 \\ 2 & 1 & 0 & -1 \\ 2 & 2 & -1 & 1 \end{array}\right)\begin{array}{l} \\ R_2 \to R_2 - 5R_1 \\ R_3 \to R_3 - 2R_1 \\ R_4 \to R_4 - 2R_1 \end{array} \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 6 & -12 \\ 0 & -3 & 2 & -7 \\ 0 & -2 & 1 & -5 \end{array}\right)\begin{array}{l} \\ R_2 \to -R_2/6 \\ \\ \\ \end{array}$$

$$\longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & -3 & 2 & -7 \\ 0 & -2 & 1 & -5 \end{array}\right)\begin{array}{l} R_1 \to R_1 - 2R_2 \\ \\ R_3 \to 3R_2 + R_3 \\ R_4 \to R_4 + 2R_2 \end{array}$$

$$\longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & -1 \end{array}\right)\begin{array}{l} R_1 \to R_1 + R_3 \\ R_2 \to R_2 - R_3 \\ \\ R_4 \to R_4 - R_3 \end{array}$$

$$\longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right)\begin{array}{l} \\ \\ R_3 \to -R_3 \\ \\ \end{array}$$

$$\longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

It follows that the unique solution is $(C_1, C_2, C_3) = (-2, 3, 1)$ and

$$\mathbf{v} = -2\mathbf{u}_1 + 3\mathbf{u}_2 + \mathbf{u}_3.$$

9. Let $\mathbf{u}$ and $\mathbf{v}$ be two nonparallel nonzero vectors with three components. Show that the vectors $\mathbf{u}$,$\mathbf{v}$ and $\mathbf{u} \times \mathbf{v}$ are linearly independent.

Solution: Since the vectors $\mathbf{u}$ and $\mathbf{v}$ are not parallel, there is a unique plane π passing through the origin $(0, 0, 0)$ and containing the two vectors. We know that the cross product $\mathbf{u} \times \mathbf{v}$ is orthogonal to $\mathbf{u}$ and to $\mathbf{v}$. In particular none of these vectors are parallel. Hence, the only possibility for the three vectors to be linearly dependent is if they are coplanar (all lie in the same plane). But this is impossible since $\mathbf{u} \times \mathbf{v}$ is orthogonal to π.

10. Consider the vectors

$$\mathbf{u}_1 = \langle 1, 2, 0, 0 \rangle, \qquad \mathbf{u}_2 = \langle -2, 1, 0, -7 \rangle, \qquad \mathbf{u}_3 = \langle 13, 0, 2, 0 \rangle, \qquad \mathbf{u}_4 = \langle 1, 0, 0, 3 \rangle.$$

Suppose that the vector $\mathbf{v} = \langle 4, 0, 1, 0 \rangle$ can be written as a linear combination

$$C_1\mathbf{u}_1 + C_2\mathbf{u}_2 + C_3\mathbf{u}_3 + C_4\mathbf{u}_4 = \mathbf{v}.$$

Use Cramer's rule to find C_2. You are not allowed to use any other method.

Solution: The numbers C_1, C_2, C_3 are C_4 form a solution of the system

$$\begin{aligned} C_1 - 2C_2 + 13C_3 + C_4 &= 4 \\ 2C_1 + C_2 &= 0 \\ 2C_3 &= 1 \\ -7C_2 + 3C_4 &= 0. \end{aligned}$$

By expanding along the third row we find that the determinant of the associated matrix is

$$\begin{aligned} \begin{vmatrix} 1 & -2 & 13 & 1 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & -7 & 0 & 3 \end{vmatrix} &= 2 \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & 0 \\ 0 & -7 & 3 \end{vmatrix} \\ &= 2 \begin{vmatrix} 1 & 0 \\ -7 & 3 \end{vmatrix} - 4 \begin{vmatrix} -2 & 1 \\ -7 & 3 \end{vmatrix} \\ &= 2. \end{aligned}$$

In particular it is nonzero and we can apply Cramer's rule. Let A_2 be the matrix obtained by replacing the second column of A by $\mathbf{v}$. By expanding along the last row we find that its determinant is

$$\begin{aligned} \begin{vmatrix} 1 & 4 & 13 & 1 \\ 2 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{vmatrix} &= 3 \begin{vmatrix} 1 & 4 & 13 \\ 2 & 0 & 0 \\ 0 & 1 & 2 \end{vmatrix} \\ &= -6 \begin{vmatrix} 4 & 13 \\ 1 & 2 \end{vmatrix} \\ &= 30. \end{aligned}$$

Hence by Cramer's rule

$$C_2 = \frac{|A_2|}{|A|} = 15.$$